Zewail City School of Science and Technology		Group: 4th Year
School of Computational Sciences and		Course Name: Reinforcement Learning
Artificial Intelligence		Course Code: DSAI 402
Date: November 10, 2025		Time Allowed: 1.5 hours
Total Marks: 20		No. of Pages: 5

Answer the questions in the spaces provided on the question sheets.

Name: _____

Serial No.: _____

Question No.	Question Description	Points	Grade
1	MRP	3	
2	MDP	6	
3	Policy Iteration	6	
4	Policy Iteration	3	
5	Value Iteration	2	
	Total	20	

In **ALL** questions, you need to show **all the steps** required to obtain your answers.

Do not answer in this page!

Q1 (3 points) Consider a simplified urban navigation problem where an agent moves between city locations with associated rewards based on traffic conditions and convenience. The locations are: Home, Bus Stop, Office, and Cafe. During a day, the agent observes the following sequences of locations and received rewards (representing satisfaction or time cost; positive is good):

- Episode 1: Home, 0, Bus Stop, -1, Office, 5, Cafe, 2
- Episode 2: Bus Stop, -2, Office, 3, Cafe, 1, Home, 3

(a) (1 point) Compute the return G_t for each state occurrence in the episodes. Assume $\gamma = 0.8$.

Episode 1: G				
Episode 2: G				

(b) (1 point) Using the returns computed, estimate the value function $v(s)$ for each state.

state	Home	Bus Stop	Cafe	Office
v				

(c) (1 point) Recompute G_t and $v(s)$ when $\gamma = 0$. Explain how the discount factor influences the value estimates in this navigation scenario.

Episode 1: G				
Episode 2: G				

state	Home	Bus Stop	Cafe	Office
v				

Q2 (6 points) You are given a Markov Decision Process (MDP) with state space $S = \{s_1, s_2\}$, action space $A = \{a_1, a_2\}$, and transition probabilities and rewards as shown in the table.

State	Action	Possible Next States	Transition Probabilities	Rewards
s_1	a_1	s_1, s_2	0.5, 0.5	5, 10
s_1	a_2	s_2	1.0	7
s_2	a_1	s_1	1.0	0
s_2	a_2	s_2	1.0	8

(a) (2 points) Define the Bellman optimality equation that relates the value function $v_\pi(s)$ to the optimal policy π^*.

(b) (2 points) Assume $v_\pi(s_1) = 50$ and $v_\pi(s_2) = 55$

i. (2 points) Outline the algorithm to extract the deterministic policy $\pi(s)$ for each state.

ii. (2 points) Compute the optimal policy $\pi(s)$ for each state; i.e. calculate $\pi(s_1)$ and $\pi(s_2)$. Assume the discount factor $\gamma = 0.9$.

	s_1	s_2
π		

Q3 (6 points) Consider an MDP with state space $S = \{s_1, s_2, s_3\}$ and two actions $A = \{a_1, a_2\}$. The transition probabilities and rewards are given as follows:

State	Action	Next State	Reward
s_1	a_1	$s_1\ (0.6),\ s_2\ (0.4)$	$3, 3$
s_1	a_2	$s_2\ (1.0)$	5
s_2	a_1	$s_2\ (0.7),\ s_3\ (0.3)$	$2, 2$
s_2	a_2	$s_1\ (0.4),\ s_3\ (0.6)$	$4, 4$
s_3	a_1	$s_3\ (1.0)$	0
s_3	a_2	$s_1\ (1.0)$	1

The discount factor is $\gamma = 0.8$. Let the initial policy be π_1 with $\pi_1(s_1) = a_1$, $\pi_1(s_2) = a_2$, $\pi_1(s_3) = a_1$.

(a) (3 points) Compute the value function $v_{\pi_1}(s)$ for each state $s \in S$ by applying the policy evaluation algorithm for one iteration. Use immediate update in v.

state	s_1	s_2	s_3
v			

(b) (3 points) Perform one policy improvement step to obtain π_2, then report the new policy mapping $\pi_2(s)$ for all $s \in S$ for one iteration.

	s_1	s_2	s_3
π			

Q4 (3 points) Compute the time complexity of the policy evaluation and policy improvement algorithms.

Q5 (2 points) You are given two versions of the value iteration algorithm for a Markov Decision Process with state space S, action space A, transition probabilities $P(s'|s,a)$, reward function $r(s,a,s')$, and discount factor γ.

Version 1:
Initialize $v_0(s) = 0$ for all $s \in S$.
For each iteration $t = 0, 1, 2, \ldots$

$$v_{t+1}(s) = \max_{a \in A} \sum_{s' \in S} P(s'|s,a)\big[r(s,a,s') + \gamma v_t(s')\big].$$

Version 2 :
Initialize $v_0(s) = 0$ for all $s \in S$.
For each iteration $t = 0, 1, 2, \ldots$ and for all $s \in S$, $a \in A$

$$Q_{t+1}(s,a) = \sum_{s' \in S} P(s'|s,a)\big[r(s,a,s') + \gamma \max_{a' \in A} Q_t(s',a')\big].$$

Then update the value function as:

$$v_{t+1}(s) = \max_{a \in A} Q_{t+1}(s,a).$$

(a) (2 points) Which version correctly implements value iteration? Justify your answer.